

3.1 Data

We evaluate our approach for six target languages: French (`fr`), Russian (`ru`), Arabic (`ar`), Chinese (`zh`), Hindi (`hi`), and Vietnamese (`vi`). These languages belong to four different language families. French, Russian, and Hindi are Indo-European languages, similar to English. Arabic, Chinese, and Vietnamese belong to Afro-Asiatic, Sino-Tibetan, and Austro-Asiatic family respectively. The choice of the six languages also reflects different training conditions depending on the amount of monolingual data. French and Russian, and Arabic can be regarded as high resource languages whereas Hindi has far less data and can be considered as low resource.

For experiments that use parallel data to initialize foreign specific parameters, we use the same datasets in the work of Lample and Conneau (2019). Specifically, we use United Nations Parallel Corpus (Ziems et al., 2016) for `en-ru`, `en-ar`, `en-zh`, and `en-fr`. We collect `en-hi` parallel data from IIT Bombay corpus (Kunchukuttan et al., 2018) and `en-vi` data from OpenSubtitles 2018.⁵ For experiments that use only monolingual data to initialize foreign parameters, instead of training word-vectors from the scratch, we use the pretrained word vectors⁶ from fastText (Bojanowski et al., 2017) to estimate word translation probabilities (Eq. 3). We align these vectors into a common space using orthogonal Procrustes (Artetxe et al., 2016; Lample et al., 2018b; Joulin et al., 2018). We only use *identical words* between the two languages as the supervised signal. We use WikiExtractor⁷ to extract raw sentences from Wikipedias as monolingual data for fine-tuning target embeddings and bilingual LMs (§2.2). We *do not* lowercase or remove accents in our data preprocessing pipeline.

We tokenize English using the provided tokenizer from pretrained models. For target languages, we use fastBPE⁸ to learn 30,000 BPE codes and 50,000 codes when transferring from BERT and RoBERTa respectively. We truncate the BPE vocabulary of foreign languages to match the size of the English vocabulary in the source models. Precisely, the size of foreign vocabulary is set to

32,000 when transferring from BERT and 50,000 when transferring from RoBERTa.

We use XNLI dataset (Conneau et al., 2018) for classification task and Universal Dependencies v2.4 (Nivre et al., 2019) for dependency parsing task. Since a language might have more than one treebank in Universal Dependencies, we use the following treebanks: `en_ewt` (English), `fr_gsd` (French), `ru_syntagrus` (Russian), `zh_gsd` (Chinese), `vi_vtb` (Vietnamese), `hi_hdtb` (Hindi), and `ar_padt` (Arabic).

Remark on BPE (Lample et al., 2018a) show that sharing subwords between languages improves alignments between embedding spaces. Wu and Dredze (2019) observe a strong correlation between the percentage of overlapping subwords and mBERT’s performances for cross-lingual zero-shot transfer. On the other hand, K et al., (2020) report in their control experiments that subword overlapping has minimal contribution cross-lingual ability of BERT. Artetxe et al., (2019) confirms in their study that a shared vocabulary is not necessary for multilingual models. In our current approach, subwords between source and target are *not* shared. A subword that is in both English and foreign vocabulary has two different embeddings.

3.2 Estimating translation probabilities

Since pretrained models operate on subword level, we need to estimate subword translation probabilities. Therefore, we subsample 2M sentence pairs from each parallel corpus and tokenize the data into subwords before running fast-align (Dyer et al., 2013).

Estimating subword translation probabilities from aligned word vectors requires an additional processing step since the provided vectors from fastText are not at subword level.⁹ We use the following approximation to obtain subword vectors: the vector e_s of subword s is the weighted average of all the aligned word vectors e_{w_i} that have s as an subword

$$e_s = \sum_{w_j: s \in w_j} \frac{p(w_j)}{n_s} e_{w_j} \quad (4)$$

where $p(w_j)$ is the unigram probability of word w_j estimated from monolingual data and $n_s = \sum_{w_j: s \in w_j} p(w_j)$.

⁹In our preliminary experiments, we learned the aligned subword vectors but it results in poor performances.

⁵<http://opus.nlpl.eu>

⁶<https://fasttext.cc/docs/en/crawl-vectors.html>

⁷<https://github.com/attardi/wikiextractor>

⁸<https://github.com/glample/fastBPE>